AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 36 · STUDENT LESSON

Capstone Expo

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Show the inputs. Trace the choice. Defend the evidence.

Present a reproducible bounded application with honest failures, support records and limitations.

Keep one habit: Report consistency → full book trace → rehearsal and handover → independent activity defence.

MAIN LESSON · SECTION 1

Can you defend the result so a peer can reproduce it?

> Show inputs, contract, validation, ranking, failures and limits together.

Your learning goals

I can defend an application using reproducible **evidence**.

I can explain failures and limits honestly.

I can demonstrate a new case without hidden help.

Recall reproducible evidence

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- What makes a project result reproducible?

Your end goal

By the end, I can defend a complete recommendation with validation, eligibility and reproducible ranking.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Improve and Challenge. Give an example and a mistake that the idea helps you avoid.

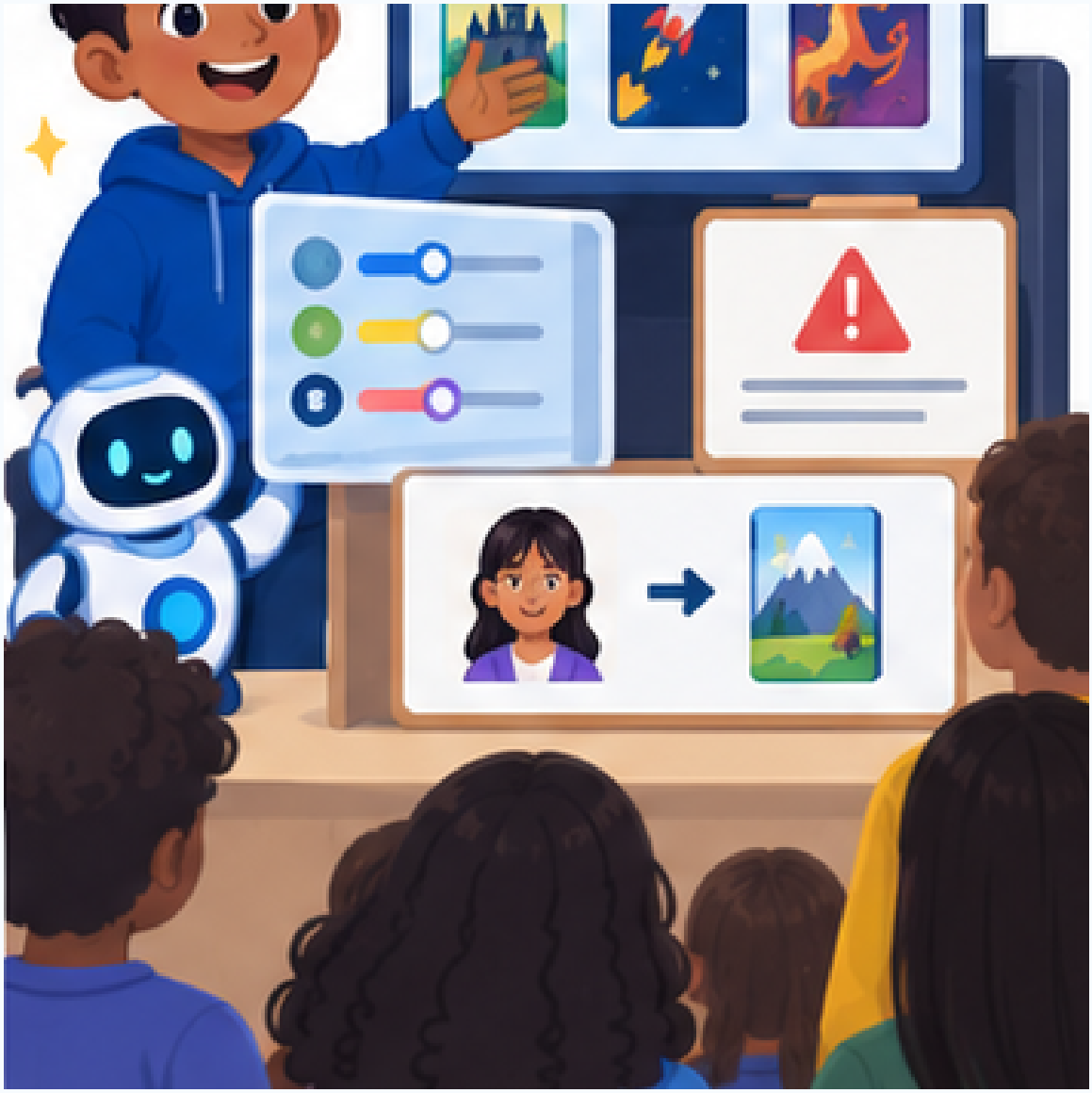
Without opening the old lesson, explain one key idea from Build the Recommender. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Agents and Tools. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 36: **Capstone** Expo

Ownership is demonstrated by a fresh recommendation and **evidence**-based **defense**.
</details>



THINKER MISSION

Demo, calculate, critique, and answer an unseen user case.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Join each demonstration claim to its evidence

> A working chooser does not by itself demonstrate educational benefit.

The capstone handover contract

Finish the fictional activity chooser with a short **evidence**-based presentation and a reproducible **handover**.

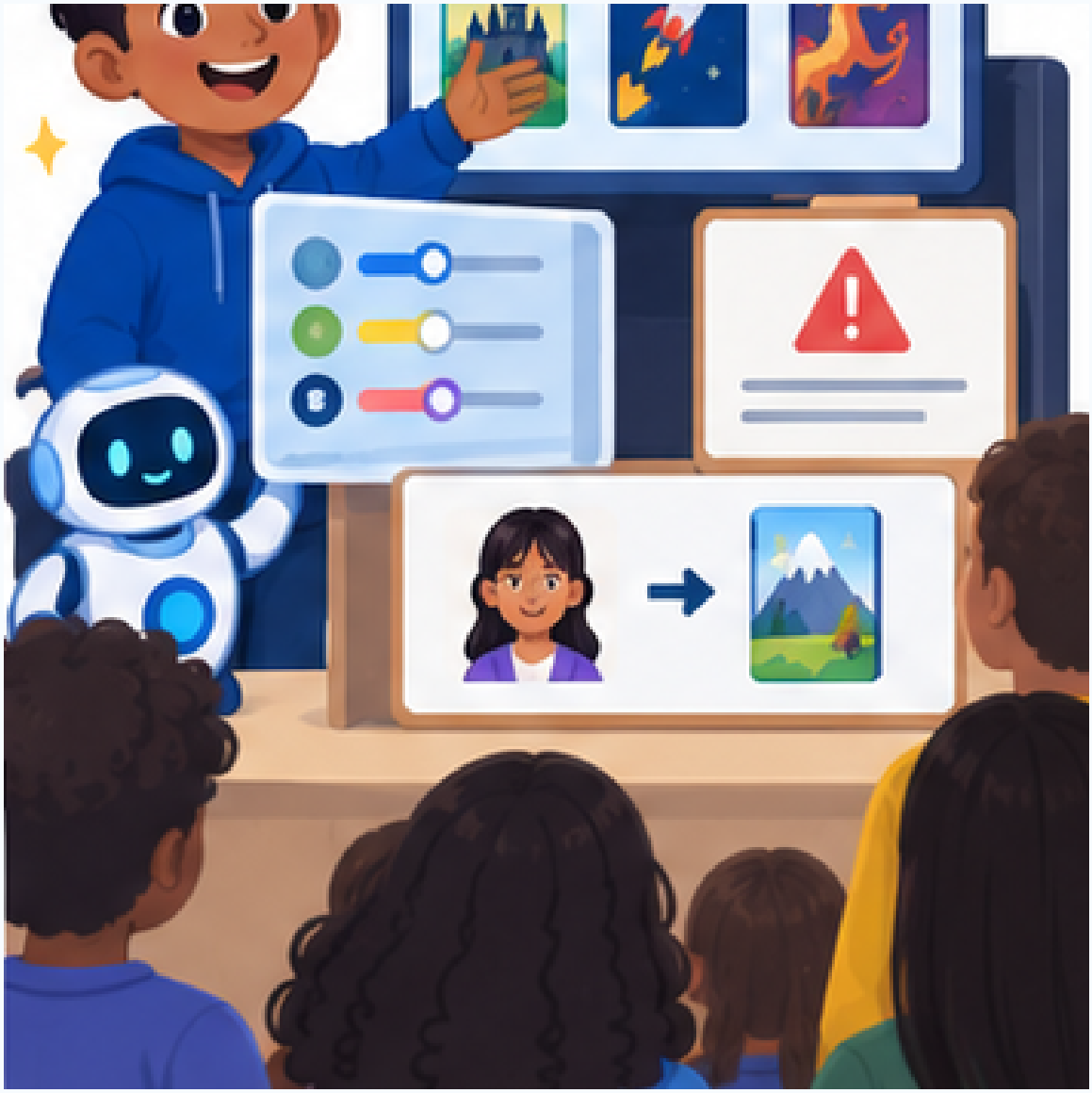
Show the purpose, contract, pipeline, one ordinary trace, one boundary or failure case, improvements and remaining limits. A working **demonstration** supports a software claim, not a claim of proven learning benefit.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use **capstone** expo to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 3

Check report consistency without mistaking it for correctness

> The small report script checks counts, not the winning recommendation.

Predict and run the complete report check

```
report = {"supplied": 4, "accepted": 3, "rejected": 1,
          "eligible": 2, "selected": "A", "score": 8}
assert report["accepted"] + report["rejected"] == report["supplied"]
assert report["eligible"] <= report["accepted"]
print(report["selected"], report["score"])
```

Trace the evidence needed beyond counts

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Claim	A selected by stated policy	Show trace and inputs
Quality	Tested cases and failures	Show full denominators
Handover	Versions and run steps	Peer can reproduce

Read each row from left to right and connect the evidence to the stated rule.

The example report prints A 8 and its counts are internally consistent. These checks do not prove A is the correct winner; the **input** records and scoring trace are also needed.

Keep raw fictional data, source code, settings, test results and change log together.

A useful **handover** joins a claim to its **evidence**. Show the fictional **input** records, the policy and version, which records passed validation, which met the time limit, the scores and any tie decision. Then show the report.

Its counts should agree, but consistent counts alone do not prove the selected item is correct.

Include an earlier failure and repair, and separate demonstrated software behaviour from educational benefits that have not been measured.

Explain the deciding evidence

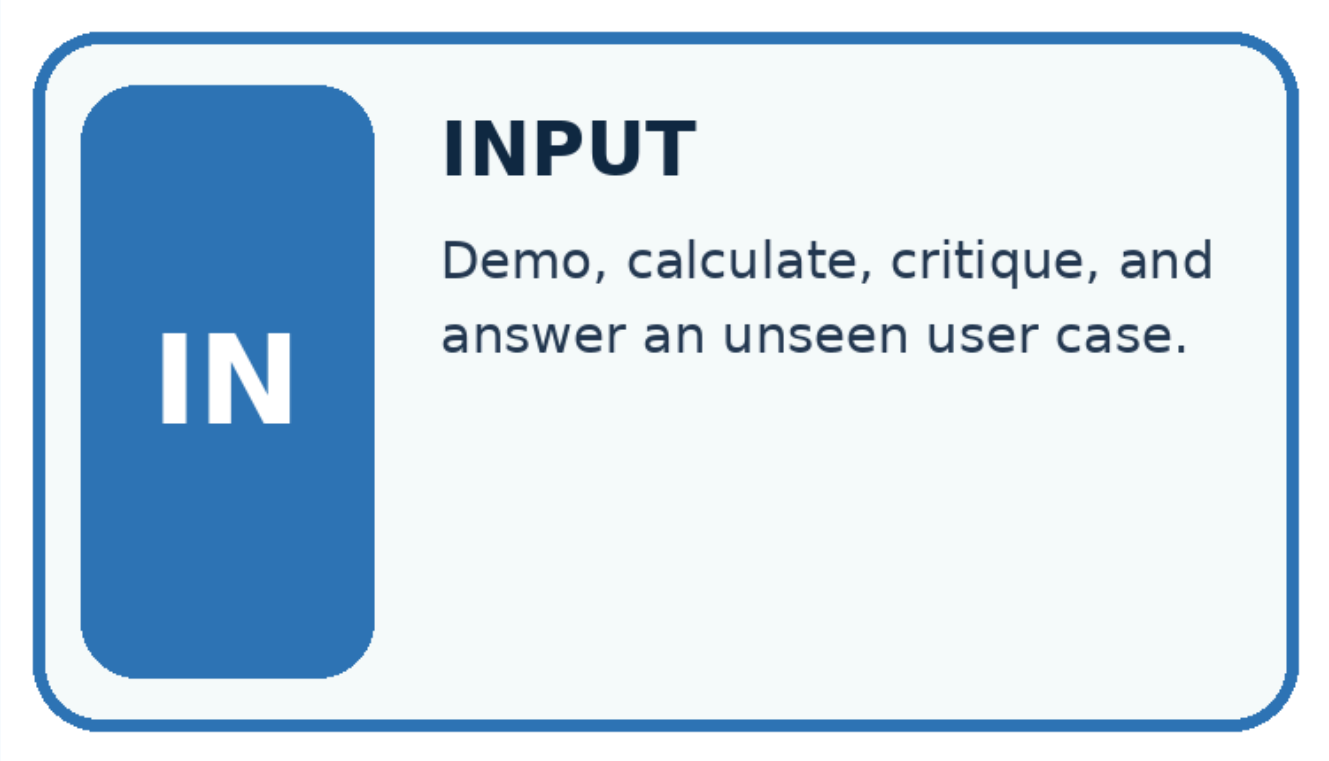
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

VISUAL **MECHANISM**

Input → **Process** → **Output**

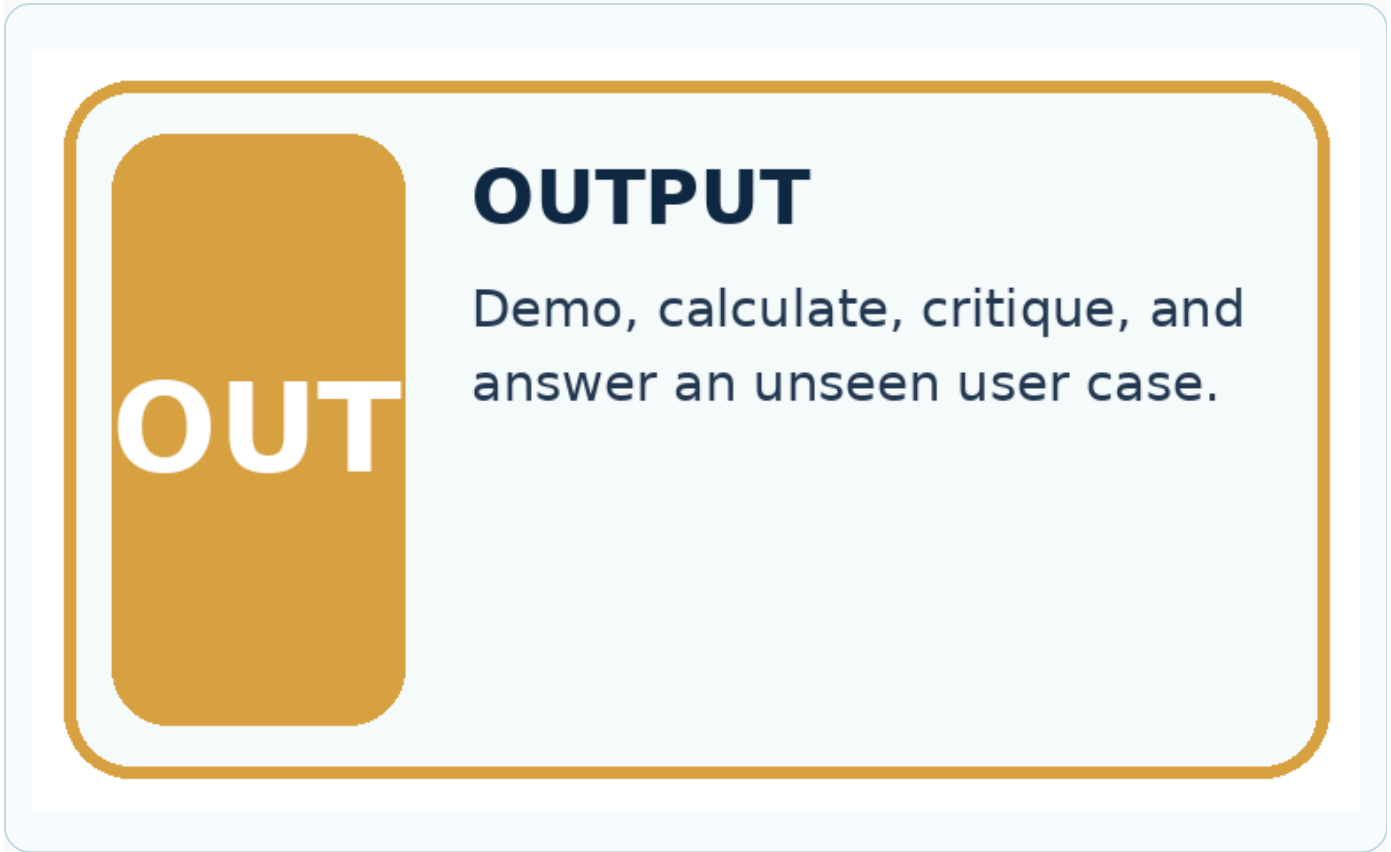
Trace the information. At each stage, ask what changed and how you could test it.



↓


↻
PROCESS
Ownership is demonstrated by a fresh recommendation and evidence-based defense.

↓


OUT
OUTPUT
Demo, calculate, critique, and answer an unseen user case.

Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

Explain a result so someone can check it

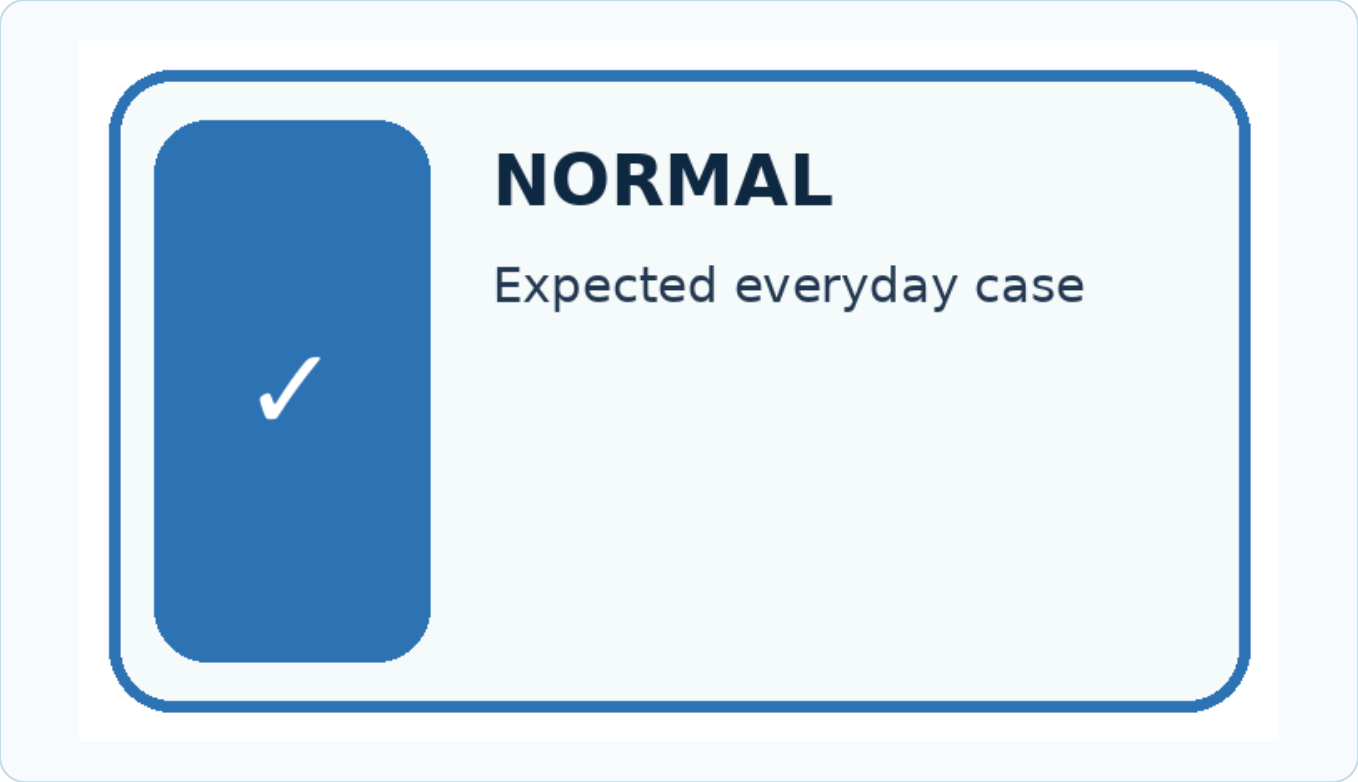
Connect the picture and the explanation

MAIN LESSON · SECTION 4

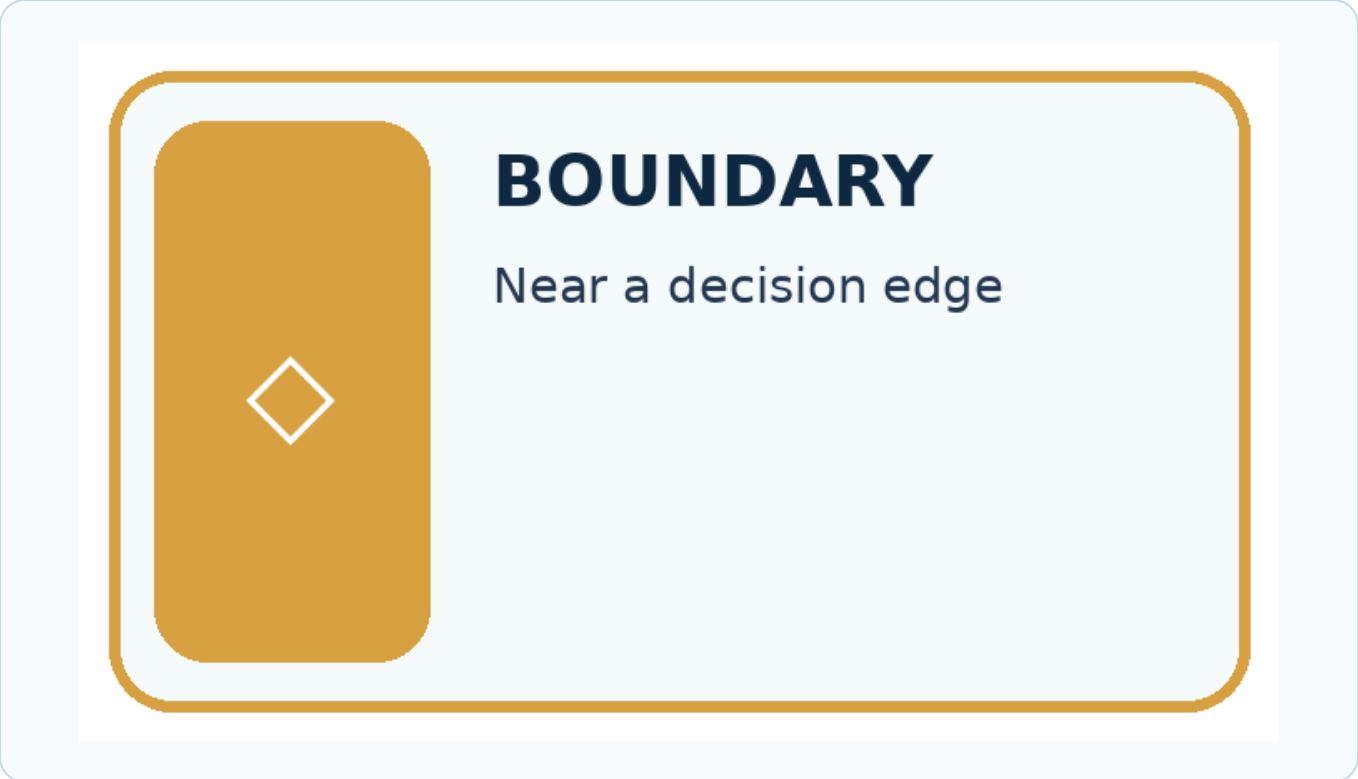
Compare a defensible trace with a polished unsupported claim

> A clear presentation keeps failures and missing evidence visible.

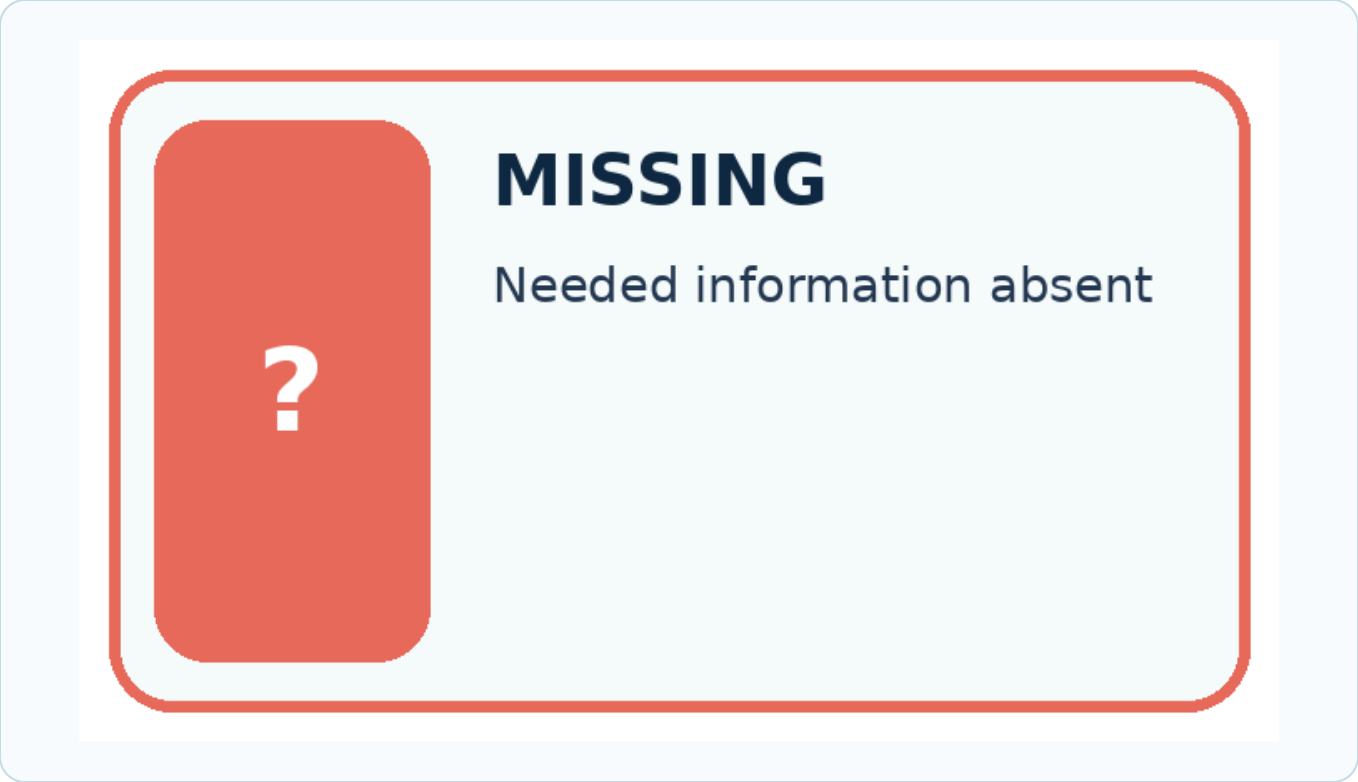
Four pictures with a purpose



Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: An authorised repair converts D's interest to integer 4.

▼ Check after predicting

Prediction: D becomes eligible and scores 9, so it wins.

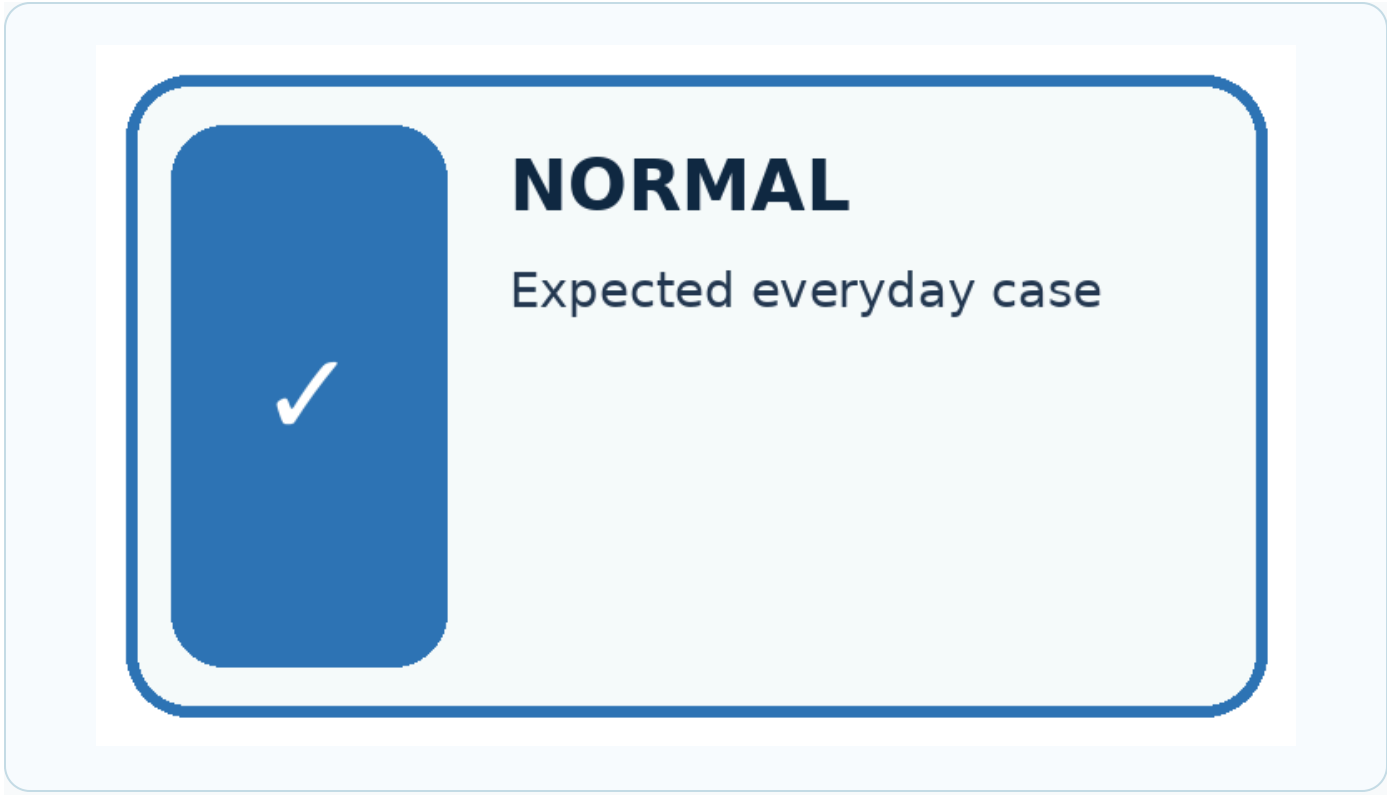
Reason: The **evidence**-backed **input** repair changes validation and then ranking; record this as a new case/version.

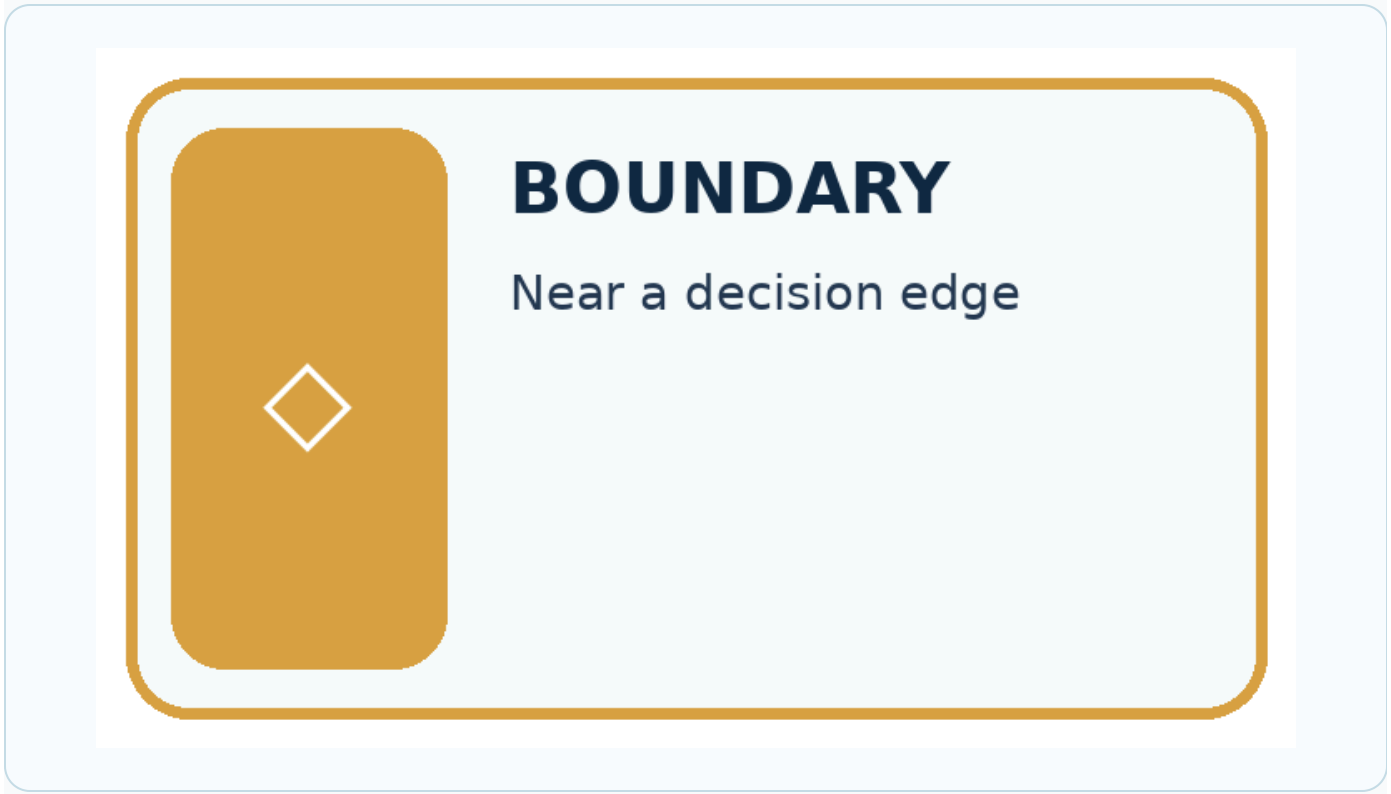
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

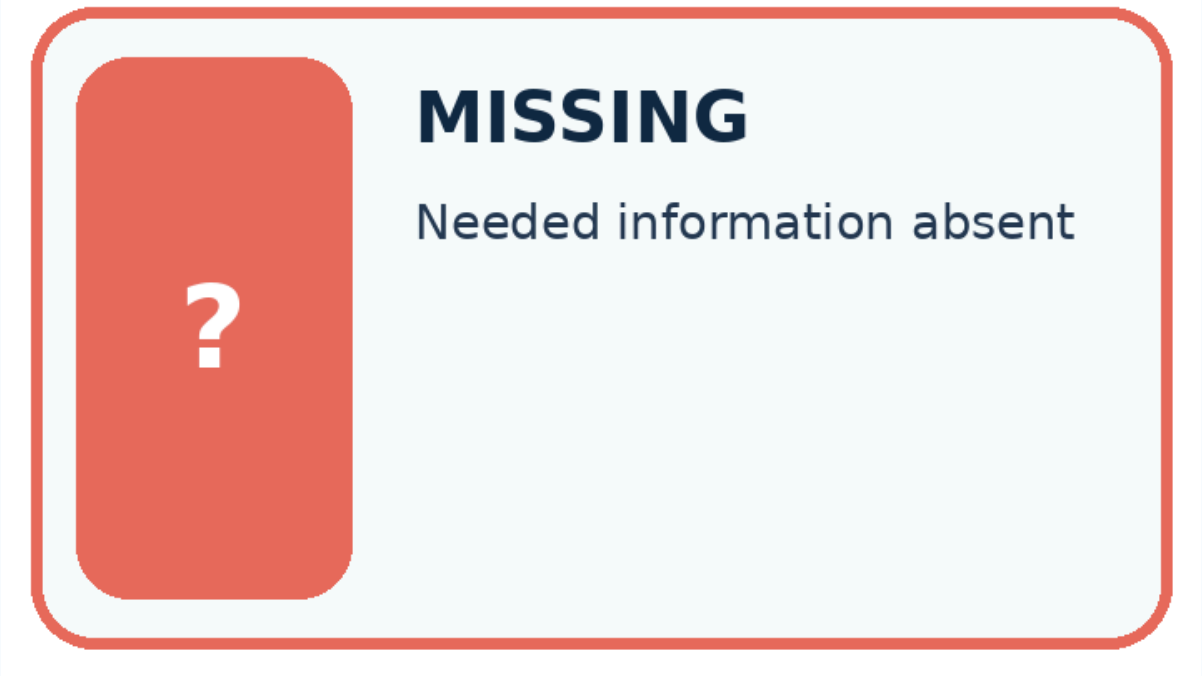
Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. **TRANSFER** CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Defend a complete book trace before presenting alone

> The pictured catalogue is supported practice, not a fresh independent assessment.

Name the evidence missing from a winner-only demo

A demonstration shows only the winning name. Name two pieces of evidence needed to explain that choice.

Read the separate book contract and rows

A fictional team presents its transparent book recommender on a supplied practice catalogue.

Request maximum 120 pages; score=2×interest+illustrated, where illustrated is 1 for True and 0 for False; tie by ID.

A pages120 interest3 True; B pages100 interest4 False; C pages121 interest5 True; D pages90 interest"4" True. Numeric fields require exact integers.

Trace validation, eligibility and ranking



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Validate	D has text interest and is rejected; A,B,C have valid fields.
2. Filter	A and B meet the page limit; C exceeds it.
3. Rank	A scores 7; B scores 8, so B wins.
4. Defend evidence	Explain two eligible items, one page-limit exclusion and one type rejection.

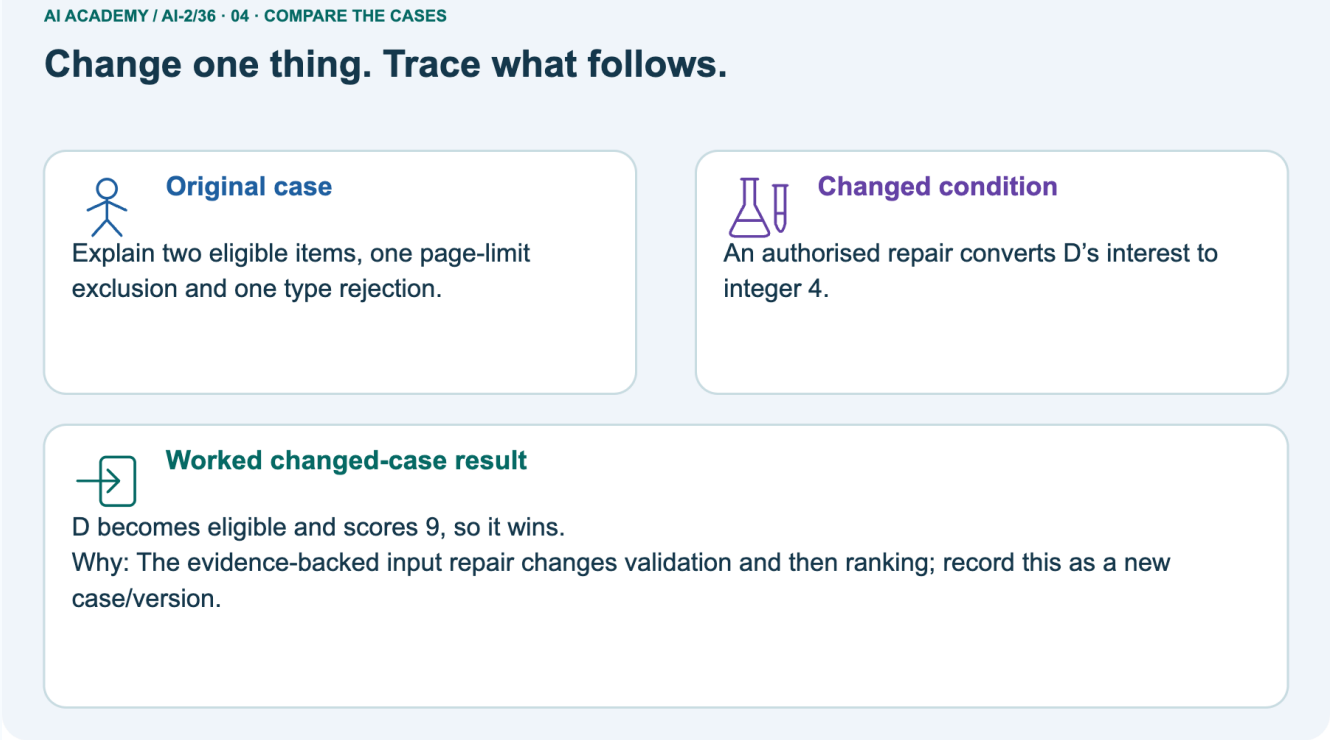
Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A presentation should let someone rebuild the result from the stated contract. A's equality is accepted; D is not silently converted; C's high preference score cannot override the page requirement. Reversing rows must preserve B because it has the unique highest eligible score. This pictured case is supported practice; keep the separately named independent activity task for a first attempt before feedback.

Predict the explicit correction to D

- Change one thing: An authorised repair converts D's interest to integer 4. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the complete implementation contract

The complete pictured implementation declares its missing schema details explicitly: nonempty unique IDs, exact integer nonnegative pages, exact integer interest0..5, Boolean illustrated; duplicate IDs are all rejected. It implements the supplied120-page limit and2interest+illustrated score with ID ties. These fields differ from the activity chooser's fun/learning schema.

The supplied correction to D is a new authorised fictional input version, not silent conversion of every text rating.

Run and inspect the complete pictured report

```
from collections import Counter

def book_report(raw, limit):
    if type(limit) is not int or limit < 0:
```

```
        return {'error':'invalid page limit'}
    names = Counter(r['id'] for r in raw if type(r) is dict
                    and type(r.get('id')) is str and r['id'])
    accepted, rejected = [], []
    for index, row in enumerate(raw,1):
        reason = None
        if type(row) is not dict:
            reason = 'not a record'
        elif type(row.get('id')) is not str or not row['id']:
            reason = 'invalid ID'
        elif names[row['id']] > 1:
            reason = 'duplicate ID'
        elif type(row.get('pages')) is not int or row['pages'] < 0:
            reason = 'invalid pages'
        elif type(row.get('interest')) is not int or not 0 <= row['interest'] <= 5:
            reason = 'invalid interest'
        elif type(row.get('illustrated')) is not bool:
            reason = 'invalid illustrated flag'
        if reason:
            rejected.append((index,reason))
        else:
            accepted.append(dict(row))
    eligible = [r for r in accepted if r['pages']<=limit]
    ranked = sorted(eligible, key=lambda r: (-
(2*r['interest']+int(r['illustrated'])),r['id']))
    scores = [(r['id'],2*r['interest']+int(r['illustrated'])) for r in ranked]
    return dict(supplied=len(raw),accepted=len(accepted),rejected=rejected,
                eligible=len(eligible),ranked=scores,
                selected=scores[0][0] if scores else None,
                over_limit=[r['id'] for r in accepted if r['pages']>limit])

books = [dict(id='A',pages=120,interest=3,illustrated=True),
         dict(id='B',pages=100,interest=4,illustrated=False),
         dict(id='C',pages=121,interest=5,illustrated=True),
         dict(id='D',pages=90,interest='4',illustrated=True)]
original = book_report(books,120)
print(original)
# Explicitly supplied authorised toy-data correction; keep the original unchanged.
changed = [dict(r,interest=4) if r['id']=='D' else dict(r) for r in books]
print(book_report(changed,120))
assert book_report(list(reversed(books)),120)['ranked']==original['ranked']
```

Explain counts and remaining limits

- A defensible algorithmic trace does not prove B will be liked or that classroom learning improved.

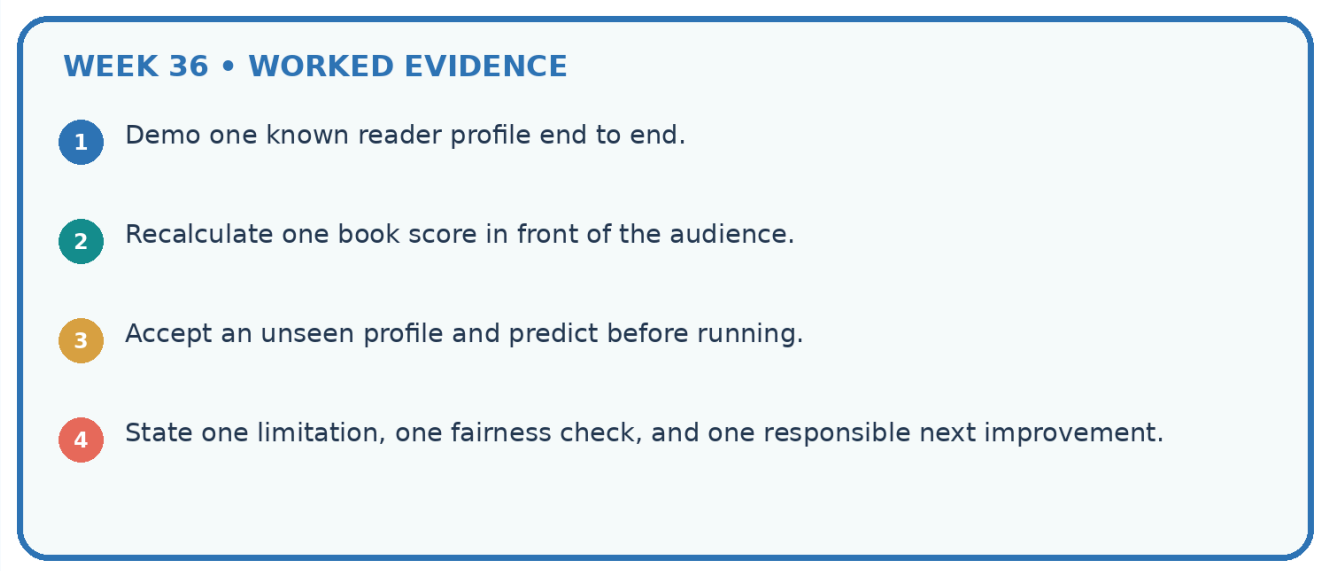
- Read the labelled picture: Give the original winning ID, its score, and counts of eligible, excluded and invalid rows.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Demo one known reader profile end to end.

2. Recalculate one book score in front of the audience.

3. Accept an unseen profile and predict before running.

4. State one **limitation**, one fairness check, and one responsible next improvement.

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

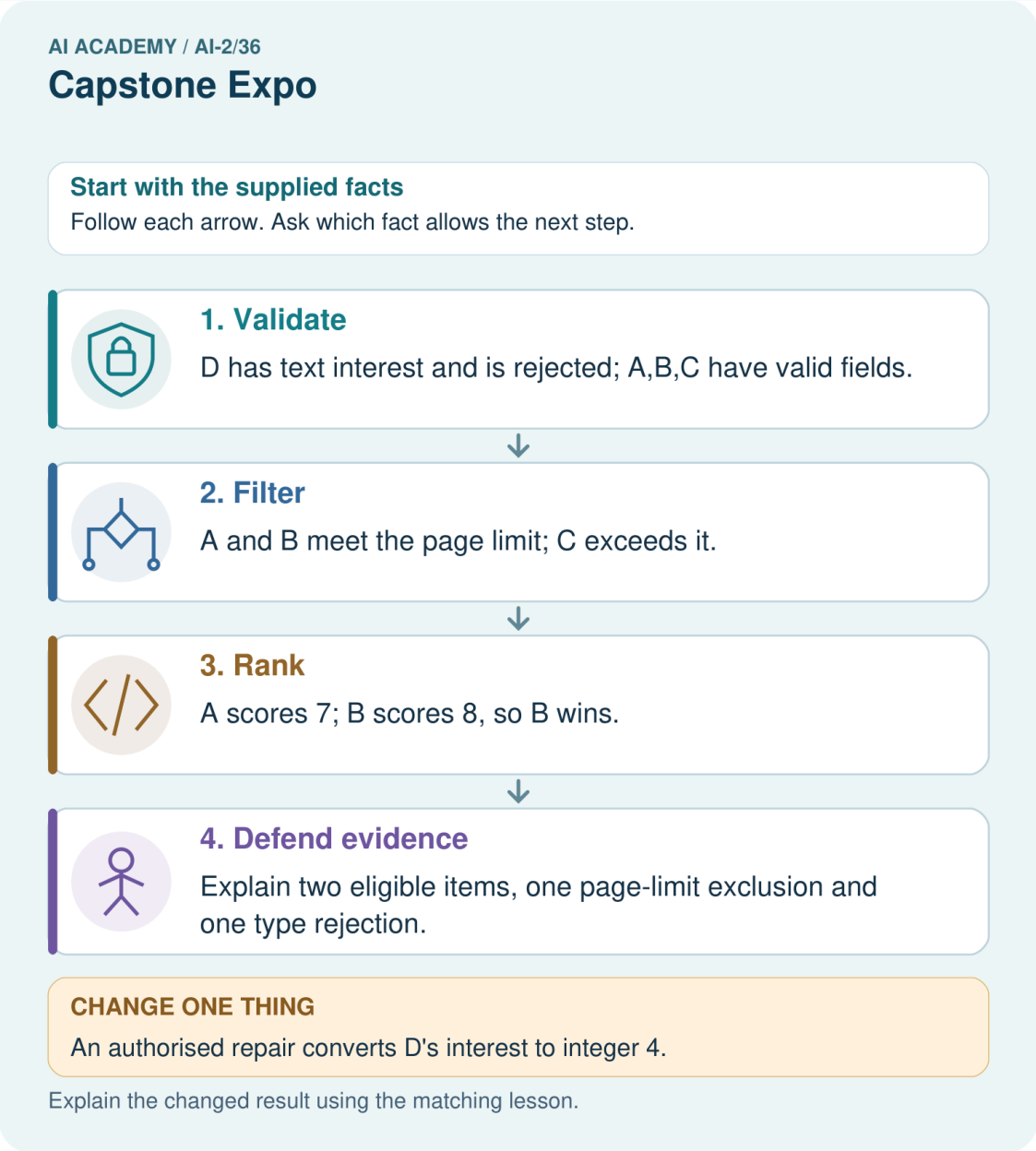
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Capstone Expo

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Challenge a correct-looking report

> Check the decision trace as well as the arithmetic of counts.

Repair the consistency argument

Claim to repair: The report’s counts add up, so A must be the correct winner. Consistent counts can accompany the wrong score or tie rule. Show raw records, schema, exact limit, score formula, each eligible score and the tie policy before validating the selected item.

The pictured book report selects B, while the small illustrative activity report says A; these are different records with different fields, not conflicting outputs for one case.

Propose a revealing audience question

After passing the new case, what remains unproven and what is a reasonable next step?

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A polished presentation is enough to prove technical understanding.
Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

The final independent task must be attempted before hints. Record support honestly and use a new case after support.

Keep teacher answers separate from student materials and do not publish private credentials or real student records in the **handover**.

MAIN LESSON · SECTION 8

Practise the evidence defence with less help

> Retain the case version, failed checks and scope of the claim.

Replace universal claims with tested behavior

A polished presentation can conceal missing **evidence**. Replace claims such as works for everyone with the exact requirements met, cases executed and known **limitation**s.

Show one prior failure and its repair rather than deleting it from the story.

Complete main W2

Practise with less help: Check report consistency

Use workbook W2 for the connected example “Defending the data application”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Repair a demo-based overclaim

- Use the worked case for Defending the data application. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Check your reasoning

Explain the separate pictured report

- Give the original winning ID, its score, and counts of eligible, excluded and invalid rows.

Explain the picture

> **My explanation using a pictured step and the given facts**
>
> | |
> |---|
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Reserve the independent activity case

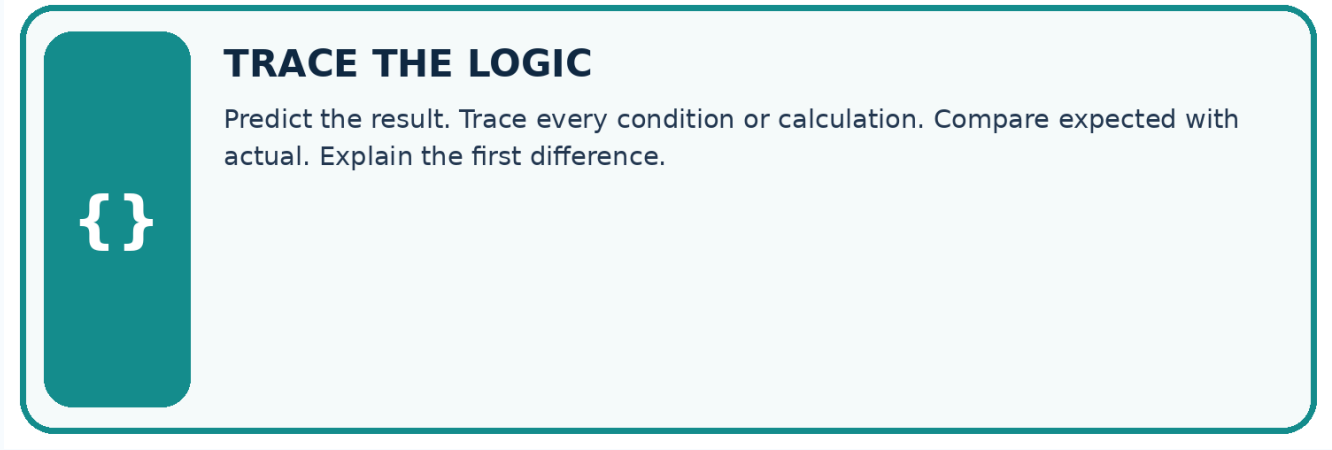
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, **limitation**, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

MAIN LESSON · SECTION 9

Rehearse a reproducible demonstration and handover

> Use supplied local data; no accounts, publication or real learner profiles are required.

Predict, run and explain the supplied report

1. Expo rehearsal: use the complete pictured book program and its supplied rows. Predict the accepted/rejected/eligible counts, scores and winning ID before running. Record actual output only if run; otherwise label a paper trace. Show the supporting row for each claim. Reverse input order, trace no eligible rows with limit0, and inspect the explicit corrected-D version without overwriting original data.

Assemble and peer-check the evidence bundle

1. Prepare a reproducible handover: purpose and non-goals; raw fictional records/version; schema; source code and settings; exact run steps; expected and observed outputs; original failure and change log; regression coverage; and remaining limits. Ask a peer to reproduce the supported case from the bundle.

   A peer question after rehearsal is not automatically a fresh assessment; the teacher must withhold a separately checked new case until the independent attempt. Do not invent pupil work or claim learning benefit from the supplied toy outcomes.

Keep the investigation record

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____

> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

PREDICT → TEST → RECORD → EXPLAIN

Present the working recommender, evidence table, test record, and limitation; answer a fresh calculation and a ‘what could fail?’ question.

Rehearse with the complete pictured book program and fictional rows, preserving expected/actual outputs and the handover bundle. Keep the independent activity task withheld for a first attempt. A familiar rehearsal case cannot establish fresh transfer.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: defend the fresh activity report

> Use this task’s activity contract rather than the pictured book formula.

Independent records and standalone contract

Limit is 10. A has minutes 10, fun 2, learning 4; B has 9,3,2; C has 11,5,5; D has 8,"4",2. Validate, rank, give counts, explain the winner, then reverse the **input** order. Present without hints.

Activity contract for this level: nonempty unique names; minutes exactint0..60;fun/learning exactint0..5; Boolean not integer; validate before selecting; eligibleminutes<=limit; score2fun+learning; highest score then ascending name; invalid rows retained with reasons; distinguish invalid request and no eligible option. Pictured book program uses different fields and score2interest+illustrated with ID ties.

Your first attempt and help record

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
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> | |

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh **transfer**.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Defend a new case independently

Week 36 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Your demo recommends the right book once. An audience member asks why they should trust it. Name **evidence** and one **limitation** you should present.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Final **evidence bundle**: brief, data, scoring logic, tests, improvement record, and product card.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Present a recommender that gets 8/10 final requests right. One failure invents a title; another violates a hard page limit. Is '**80%**, ready for everyone' justified? Propose release criteria.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Finish a project bundle someone else can check

> A defensible handover includes the failures and evidence limits.

This week’s concrete contribution

Package the Transparent Book Recommender’s own schema, code, data, source-supported facts, test matrix, full ranking traces, no-match behavior, repair history and limitations. Label supplied classroom examples separately from work a learner actually completed. Provide local reproduction steps and record support used.

A final showcase assesses the explanation and evidence for this bounded project; revisit other level concepts separately where readiness evidence is weak.

Present the Transparent Book Recommender’s code, baseline, full-case traces, reserved results and **limitation**s, without claiming fictional results as pupil work.

Keep the checkpoint and human choice

THIS WEEK’S CHECKPOINT Final **evidence bundle**: brief, data, scoring logic, tests, improvement record, and product card.

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

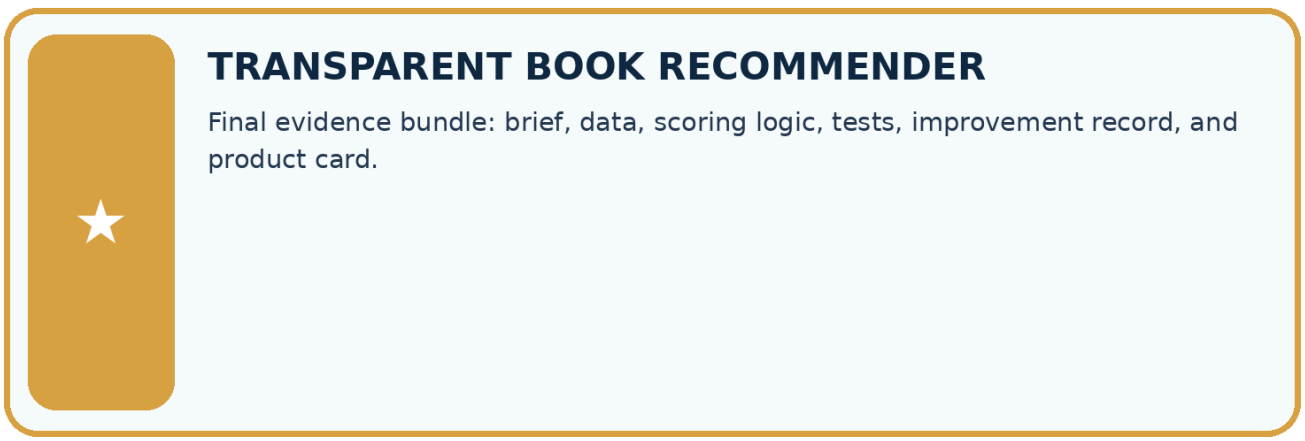
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse a ranking with a fact about a person.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: defend the decision and name the next evidence need

> A completed expo does not establish mastery of every AI topic.

Explain without notes

- Close your notes. Explain this goal in your own words: Defend an application using reproducible **evidence**

> **My deciding step, evidence and one remaining question**

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
capstone	A final project combining skills and evidence developed across the course.	
defense	An explanation of a project decision supported by evidence and an account of its limitations.	
transfer	Apply an idea or skill to a new case rather than copy the worked example.	
limitation	A condition or missing capability that restricts what a result can establish or a system can safely do.	

Terms for the connected explanation

Word	Meaning
Evidence bundle	The records needed to check a project claim
Demonstration	Showing the application on specified inputs

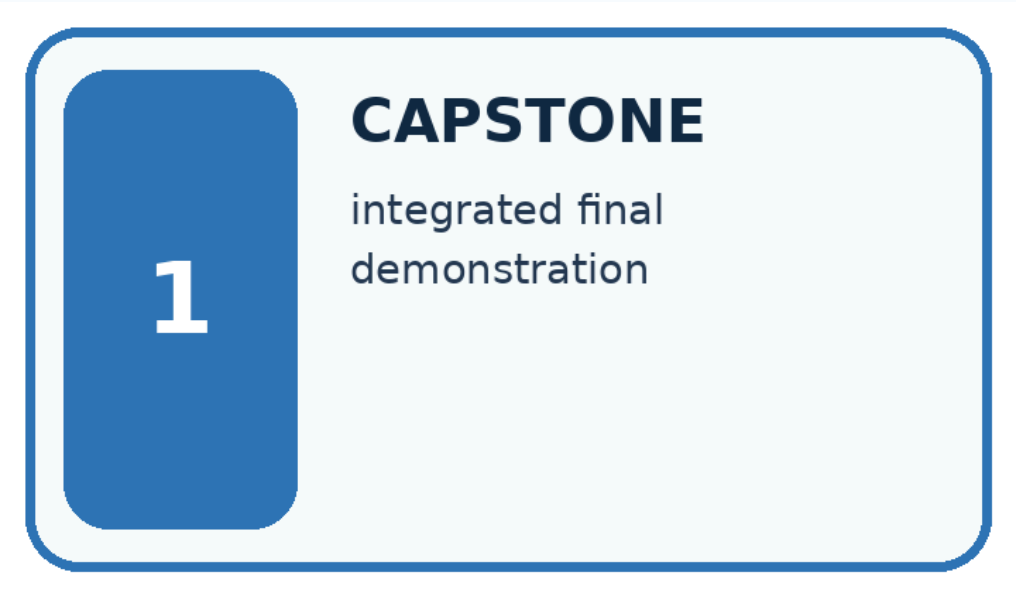
Word	Meaning
Handover	Giving another person enough information to use or check the work

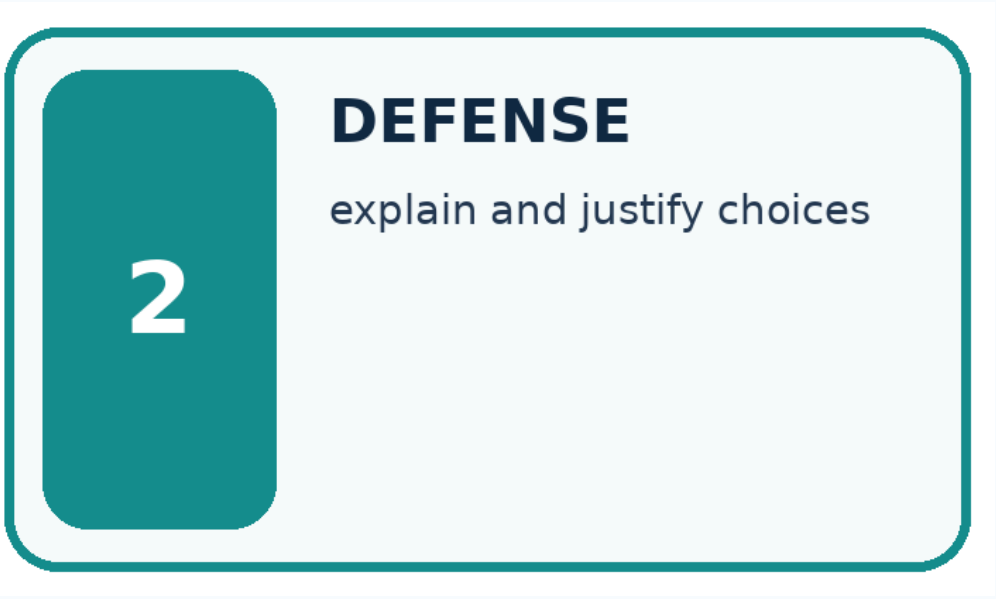
▼ Earlier wording, alternatives and picture transcripts

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
DEFENSE
explain and justify choices


3
TRANSFER
apply learning to a new case



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

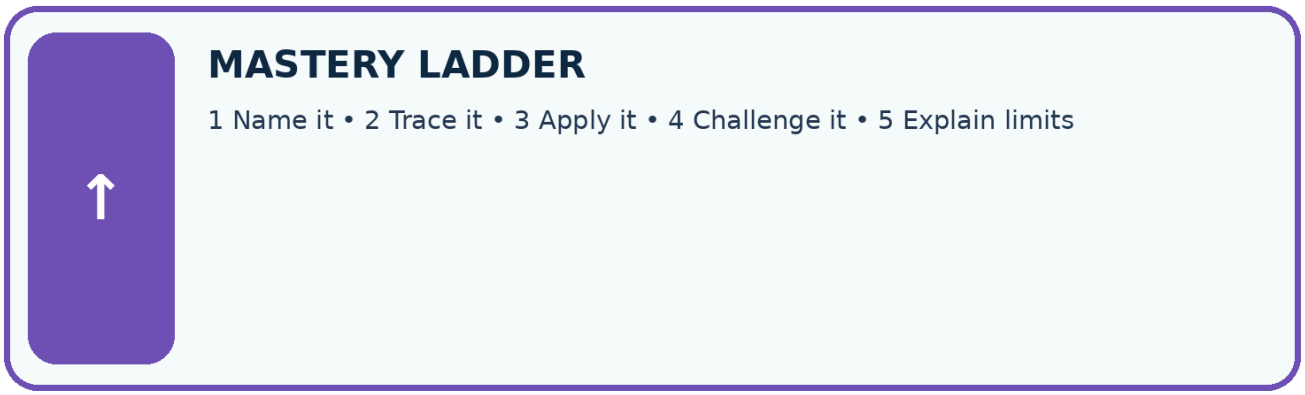
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Defend a new activity case independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One **limitation**, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

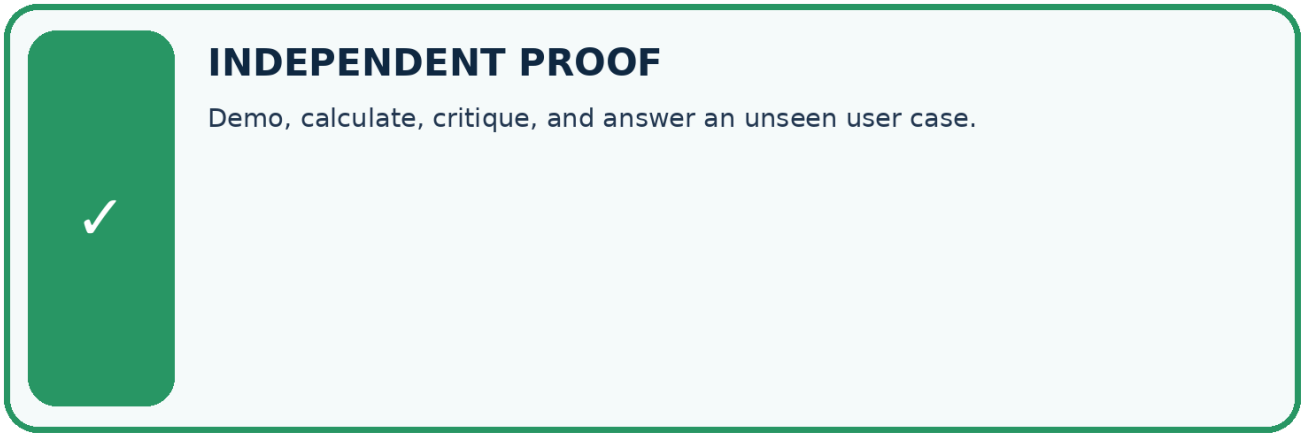
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a **limitation** ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Defend a new activity case independently is the selected independent case.



EXPLANATION CHECK Require a complete schema, filtering/score/tie trace, counts, reversed-order result and evidence limits.

Write or draw your reasoning and evidence.

▼ **Extra practice: Defend a transparent recommender**

REFERENCE · SECTION 13

Extra practice: Defend a transparent recommender

Try the question before opening its worked explanation.

Read the supplied case and question

An expo claim says the recommender knows everybody's favourite book after twelve tests.

Rewrite the defence.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Explain trait-based ranking, report the twelve outcomes and limits, and demonstrate no-match handling. It cannot know unstated preferences.

Evidence or stage	Value or result
Inputs	Explain necessary data
Logic	Show a score and tie decision
Tests	Report successes and failures
Limits	Demonstrate a missing or uncertain case

▼ Earlier wording, alternatives and picture transcripts

Week 36 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Defend a transparent recommender

An expo claim says the recommender knows everybody's favourite book after twelve tests.

Rewrite the defence.

Explain trait-based ranking, report the twelve outcomes and limits, and demonstrate no-match handling. It cannot know unstated preferences.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Defend an application using reproducible evidence	Explain the deciding step, show a trace or test, and state one limit.
Explain failures and limits honestly	Explain the deciding step, show a trace or test, and state one limit.
Demonstrate a new case without hidden help	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional team presents its transparent book recommender on a supplied practice catalogue.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional team presents its transparent book recommender on a supplied practice catalogue.
- Which labelled step matches this detail? Explain two eligible items, one page-limit exclusion and one type rejection.
- 1. Defend **evidence**
- 2. Validate
- 3. Filter
- 4. Rank
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Defend **evidence** Explain two eligible items, one page-limit exclusion and one type rejection. A presentation should let someone rebuild the result from the stated contract. A's equality is accepted; D is not silently converted; C's high preference score cannot override the page requirement. Reversing rows must preserve B because it has the unique highest eligible score. This pictured case is supported practice; keep the separately named independent activity task for a first attempt before feedback.

Changed-case question

- Changed-case practice: An authorised repair converts D's interest to integer 4. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. D becomes eligible and scores 9, so it wins.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: D becomes eligible and scores 9, so it wins. The **evidence**-backed **input** repair changes validation and then ranking; record this as a new case/version.

Repair and save your work

Review Improve and Challenge (ai-2/35). Goal: Repair a specific observed failure. Short practice: Why are tests at both 100 and 101 needed?. Return to this week and explain what you changed.

Review Build the Recommender (ai-2/33). Goal: Implement a deterministic recommender. Short practice: Explain why B does not win after D is excluded even though it has fewer pages..

Return to this week and explain what you changed.

Review Agents and Tools (ai-2/30). Goal: Explain the **mechanism** and terminology of state machines for bounded agents. Short practice: Can the assistant accept Trace B because its final state is done?.

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

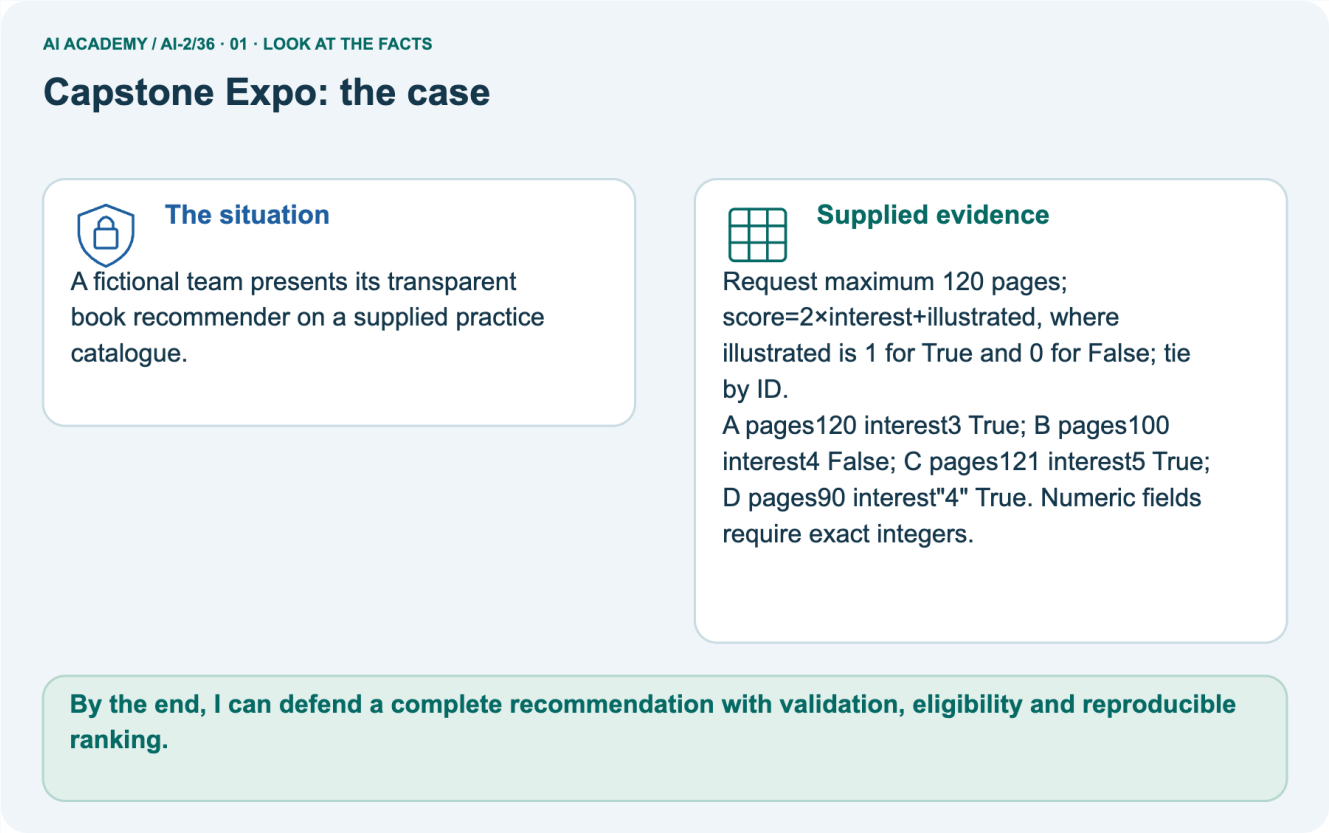
▼ **Reference: the six-picture journey**

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Capstone Expo: the case

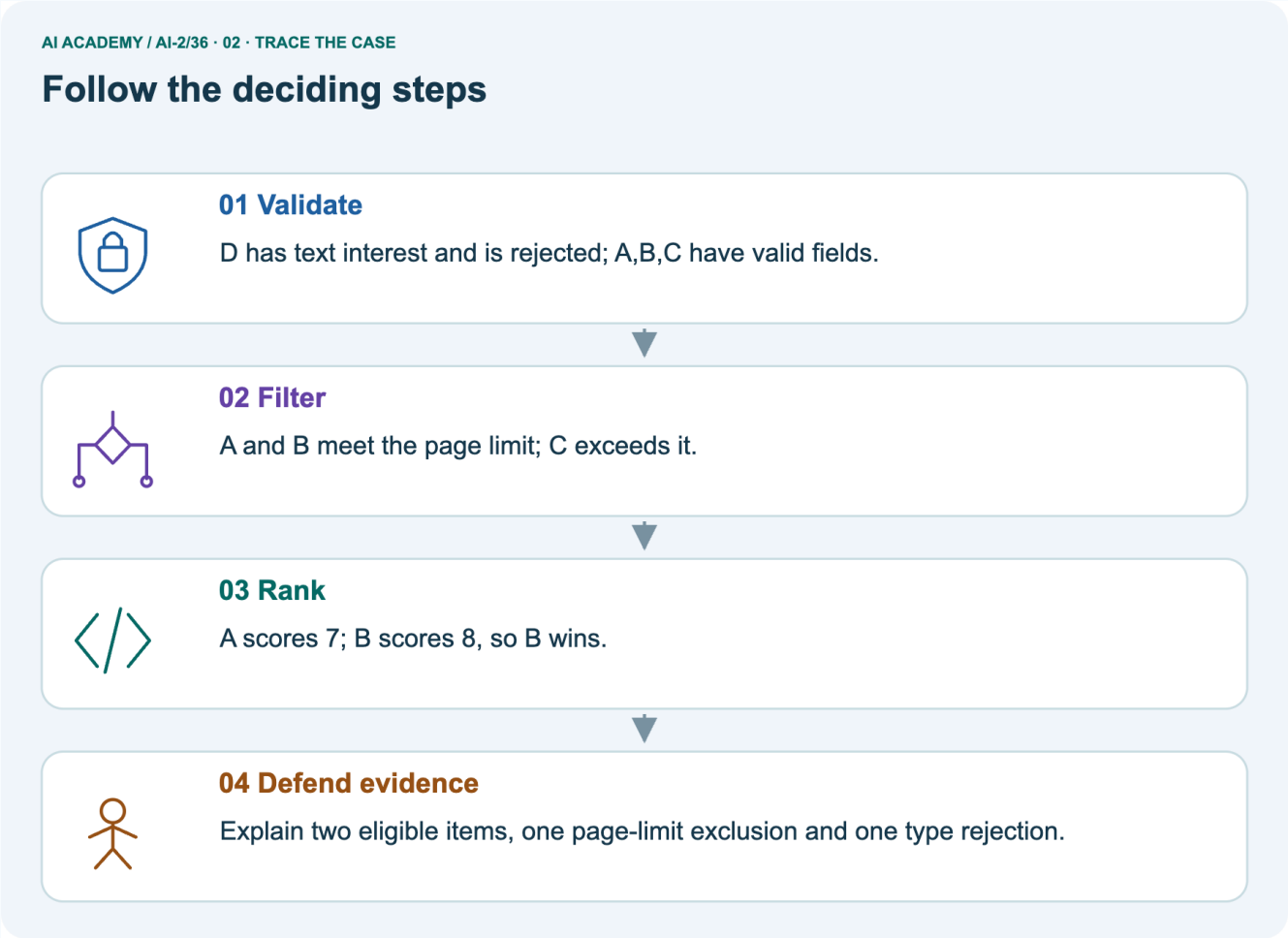


Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional team presents its transparent book recommender on a supplied practice catalogue. Supplied facts: Request maximum 120 pages; score=2×interest+illustrated, where illustrated is 1 for True and 0 for False; tie by ID. A pages120 interest3 True; B pages100 interest4 False; C pages121 interest5 True; D pages90 interest"4" True. Numeric fields require exact integers.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

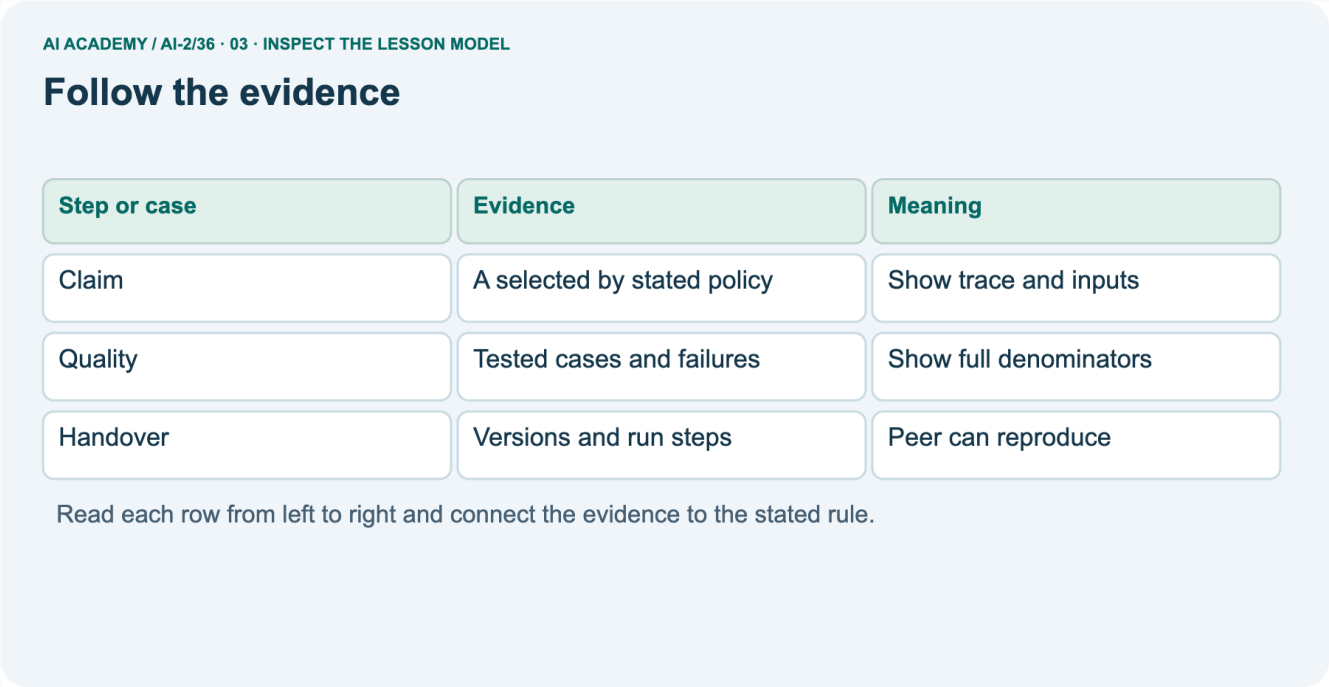


Follow the labelled picture with the matching explanation. Open the vector diagram

Validate: D has text interest and is rejected; A,B,C have valid fields. → Filter: A and B meet the page limit; C exceeds it. → Rank: A scores 7; B scores 8, so B wins. → Defend **evidence**: Explain two eligible items, one page-limit exclusion and one type rejection.

Point to each numbered stage and explain which recorded **input** it uses.

03. Follow the evidence

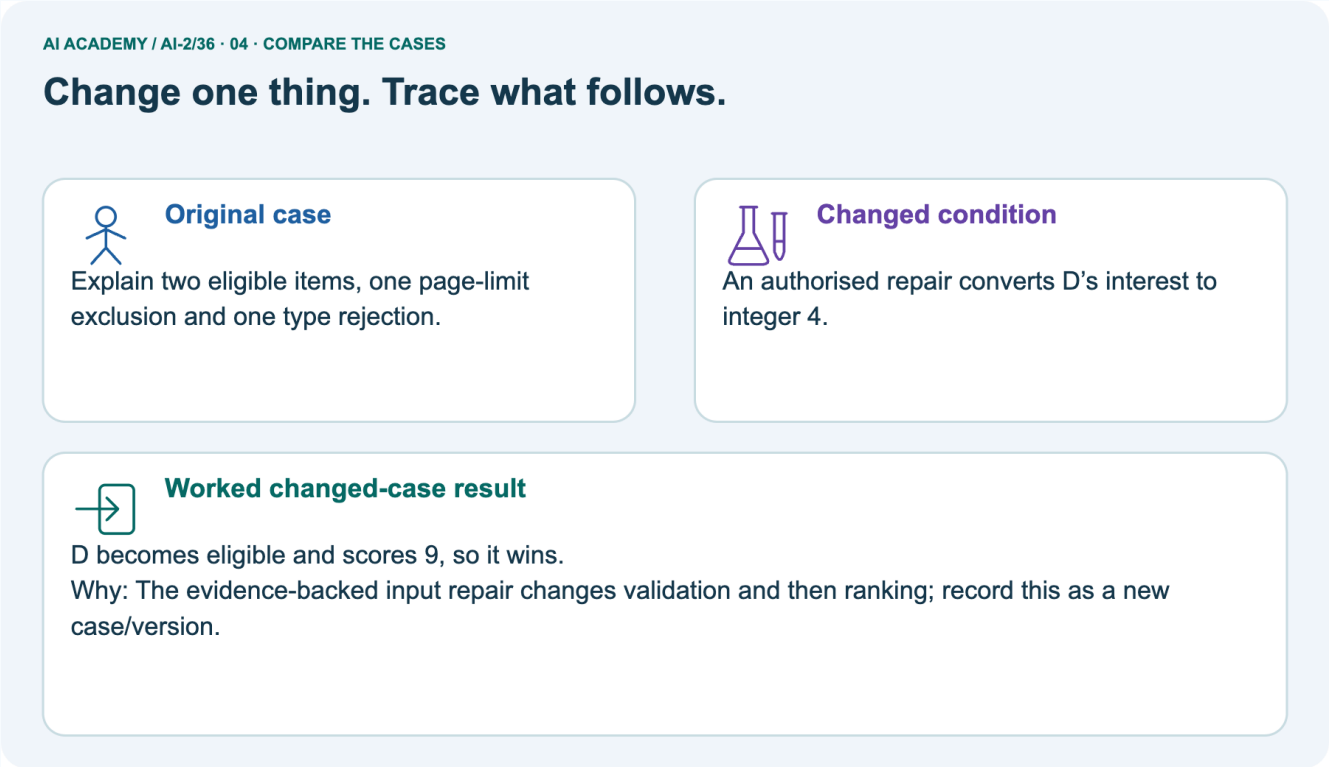


Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. Claim | A selected by stated policy | Show trace and **input**s. Quality | Tested cases and failures | Show full denominators. **Handover** | Versions and run steps | Peer can reproduce

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



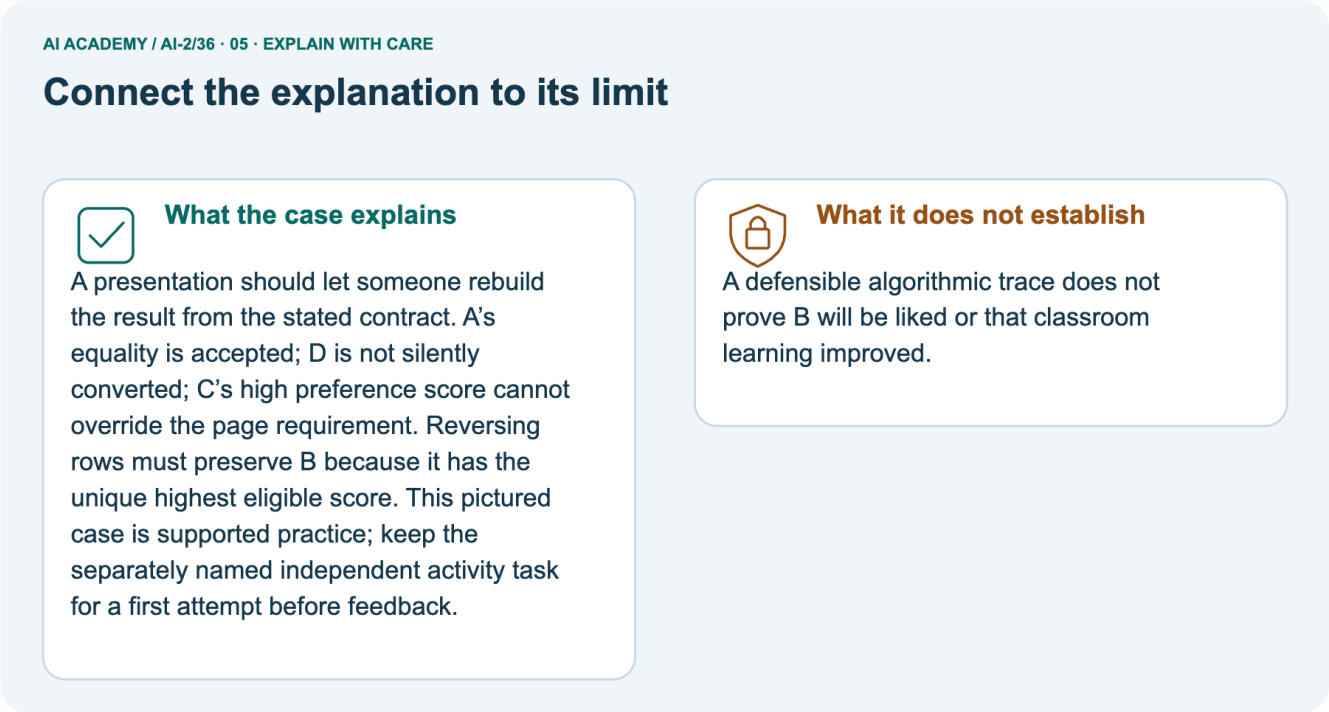
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Explain two eligible items, one page-limit exclusion and one type rejection.
Changed condition: An authorised repair converts D's interest to integer 4.

Expected result: D becomes eligible and scores 9, so it wins. Reason: The **evidence**-backed **input** repair changes validation and then ranking; record this as a new case/version.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit

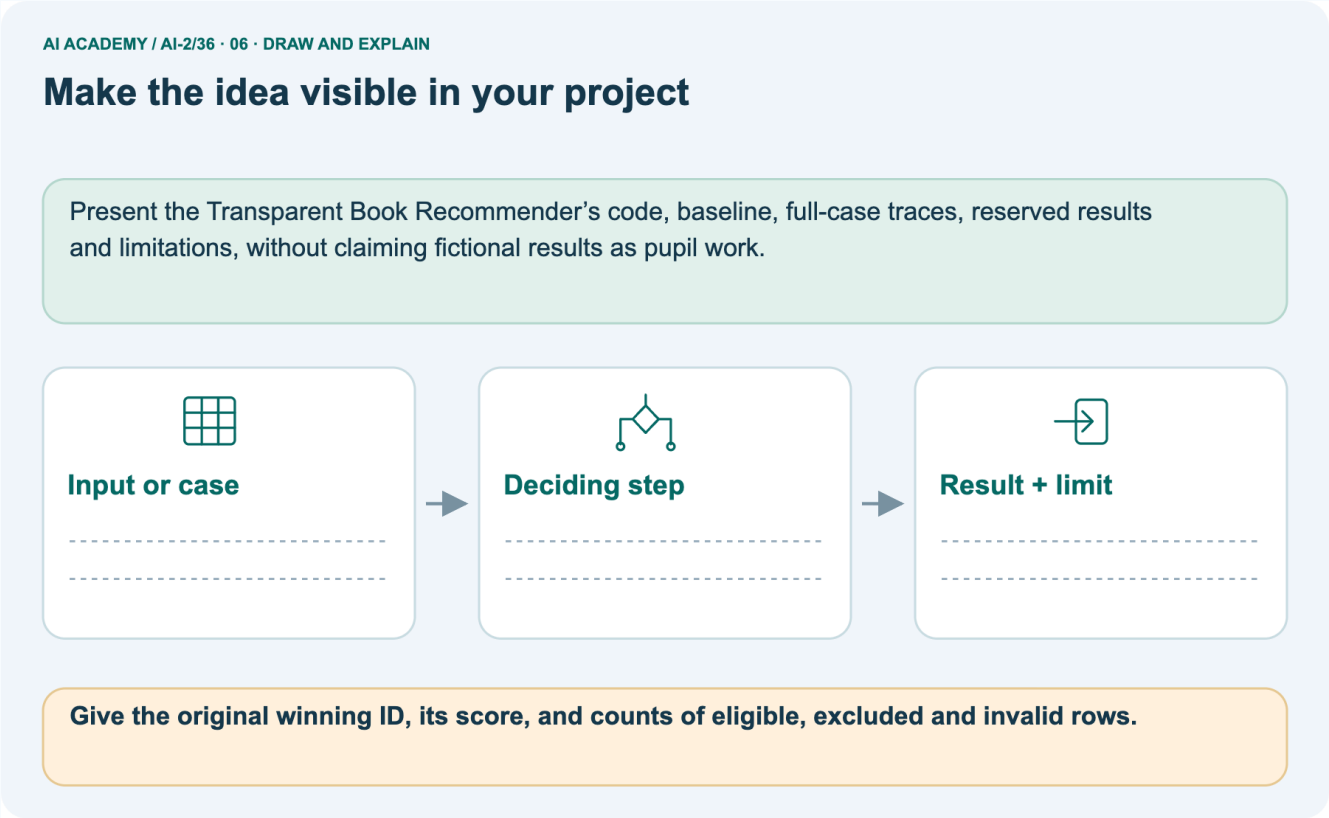


Follow the labelled picture with the matching explanation. Open the vector diagram

A presentation should let someone rebuild the result from the stated contract. A's equality is accepted; D is not silently converted; C's high preference score cannot override the page requirement. Reversing rows must preserve B because it has the unique highest eligible score. This pictured case is supported practice; keep the separately named independent activity task for a first attempt before feedback. Limit: A defensible algorithmic trace does not prove B will be liked or that classroom learning improved.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Present the Transparent Book Recommender's code, baseline, full-case traces, reserved results and **limitation**s, without claiming fictional results as pupil work. Draw your **input**, deciding step, result and limit.

Supported question: Give the original winning ID, its score, and counts of eligible, excluded and invalid rows.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Capstone Expo: the case

02. Follow the deciding steps

03. Follow the evidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.